SIGNAL

Strategic Intelligence Gateway for Next-Gen Analytics & Lending

High-Level Project Overview

Prepared for: SIGNAL project record — Team 2 (SAIT PROJ 407)
Industry partner: ATB Financial (advisory / consultation context)
Audience: New stakeholders, reviewers, and anyone needing a one-read understanding of SIGNAL
Companion documents: Doc 2 — Mid-Level Description · Doc 3 — Low-Level Detail · Doc 4 — Glossary of Terms

The One-Paragraph Version

SIGNAL is a proof-of-concept, AI-assisted decision-support system for commercial credit monitoring. It ingests borrower financial documents in whatever format they arrive: PDFs, spreadsheets, images, scanned statements, emails, validates and standardizes the data, calculates the same financial ratios a credit analyst would calculate by hand, and checks them against a bank's own Early Warning Indicator (EWI) rules. When something looks off, such as a covenant breach, a late filing, or a revenue decline, SIGNAL raises a flagged, explainable alert and shows the analyst exactly why. It never approves, declines, rates, or predicts anything on its own. Every decision stays with a human.

How SIGNAL Adds Value

Question / Metric	Value Proposition / Impact
What's the big picture?	SIGNAL is a fully AI-assisted commercial credit monitoring platform
Why is it worth doing?	Operational efficiency, governance, earlier risk detection, increased analyst capacity
How much money does it save?	Estimated \$400K–\$550K per annum in productivity improvements
How much time does it save?	60–70% time savings: From 7–8 hours manual, down to 2.5 - 3 hours
How many more loans/reviews?	50–60% increase in loan review volume

What does increased throughput mean?	More borrower reviews, more proactive engagement, better portfolio monitoring, less loan losses.
What happens next?	Enterprise integration, continuous monitoring, explainable AI.

The Problem

Commercial Portfolio Managers and Credit Analysts are responsible for continuously monitoring the financial health of every borrower in their portfolio. Each quarter, businesses submit interim financial statements, and someone has to read them, calculate the relevant ratios, compare those ratios to covenant thresholds and internal risk policy, and decide whether anything looks like an early sign of trouble.

That process works, but it does not scale. As portfolios grow, so does the manual workload. And manual, judgment-based review is inherently inconsistent: two analysts under different time pressure can review the same file and reach different conclusions about what deserves attention. Warning signs that are caught late leave the institution and the borrower with fewer options. And because the review is manual, it's hard to prove after the fact that it was done consistently and thoroughly. This is a real concern for internal governance and regulatory expectations.

> **In one sentence**
> Credit teams need a scalable, consistent, explainable way to catch early warning signs of borrower financial distress — without losing the human judgment that credit decisions ultimately require.

What SIGNAL Does

SIGNAL follows the same basic path a human analyst follows, automated end to end, with every step visible and explainable:

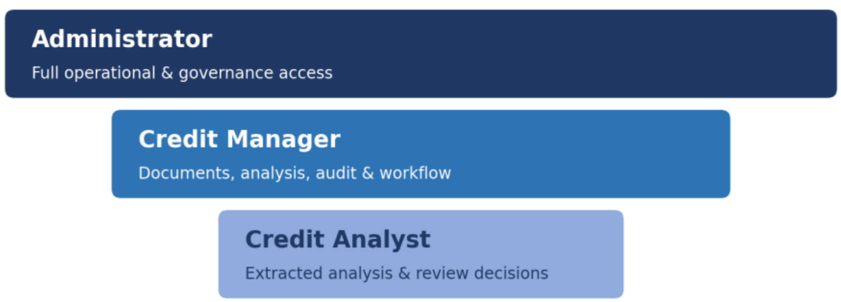
- Ingest — accept borrower financial packages in mixed formats (PDF, Excel, CSV, images, text/email).
- Extract — pull out the numbers and terms that matter: revenue, EBITDA, debt, working capital, covenant terms, key dates.
- Validate — check the extracted data for completeness, consistency, and plausibility before it's trusted.

- Analyze — calculate standard credit-monitoring ratios (leverage, liquidity, coverage, cash conversion, and more) and test them against the bank's own Early Warning Indicator rules and covenant thresholds.
- Alert — surface anything that crosses a threshold as a rated (Low / Medium / High) alert, with the source evidence and the specific rule that triggered it attached.
- Present — put all of this in front of the right person, in a role-appropriate dashboard view, so they can review, validate, correct, or act on it.

Every one of those steps is logged. Every alert traces back to a source document, an extracted value, and a named rule version, so the output can always be explained and defended, not just trusted.

Who It's For

SIGNAL is a role-based application: what a person can see and do is scoped to their role, and access narrows as you move from Administrator down to Credit Analyst. This reflects the current live build's sign-in role picker.



Role	What they can see and do
Administrator	Full operational and governance access: scheduling, system logs, rule configuration, user management, and everything available to the roles below.
Credit Manager	Document access, the analyst view, audit logs, rule changes, and the broader business workflow. This is the working view for a credit team lead who needs both oversight and evidence.
Credit Analyst	Sees what SIGNAL extracted from shared files and the resulting analysis, and records review decisions. Does not have direct access to the underlying source documents.

What SIGNAL Deliberately Does Not Do

This boundary is the most important design decision in the whole project, and it's intentional rather than a current limitation to be removed later:

- It does not approve or decline credit.
- It does not assign official credit ratings or risk scores.
- It does not predict borrower default.
- It does not recommend a course of action, restructuring, or remediation.
- It does not connect to live core banking systems or production data. The current build runs entirely on synthetic and public/reference data.

SIGNAL's job stops at surfacing relevant, well-evidenced information efficiently. Interpretation, risk judgment, and action remain entirely with qualified human professionals.

Project Status & Context

SIGNAL is a capstone project for the SAIT Post-Diploma Certificate in Integrated Artificial Intelligence, developed by Team 2 (Gurpreet Bajwa, Sukhpreet Singh, Neil McGuffin, Roy Aggarwal) in consultation with ATB Financial. It is explicitly a proof of concept: the goal is to demonstrate that this kind of AI-assisted monitoring is technically sound, explainable, and genuinely useful to credit teams. It is not meant to ship a production banking system.

The project has progressed through problem definition, business/functional requirements, and a full system architecture design, and is now in build: a working MVP dashboard and processing pipeline exist and are being actively developed.

Why It Matters

Audience	Why this matters to them
Analysts	Less time on manual data extraction and threshold-checking, more time on judgment and borrower engagement.
The institution	Earlier detection of distress preserves more options; a repeatable process is easier to defend to internal stakeholders and regulators; monitoring capacity scales without a proportional headcount increase.
Borrowers	Earlier, more constructive engagement before a financial situation becomes critical.
The industry	A working example of AI adopted responsibly in credit risk — human oversight, full explainability, no automated adjudication.

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